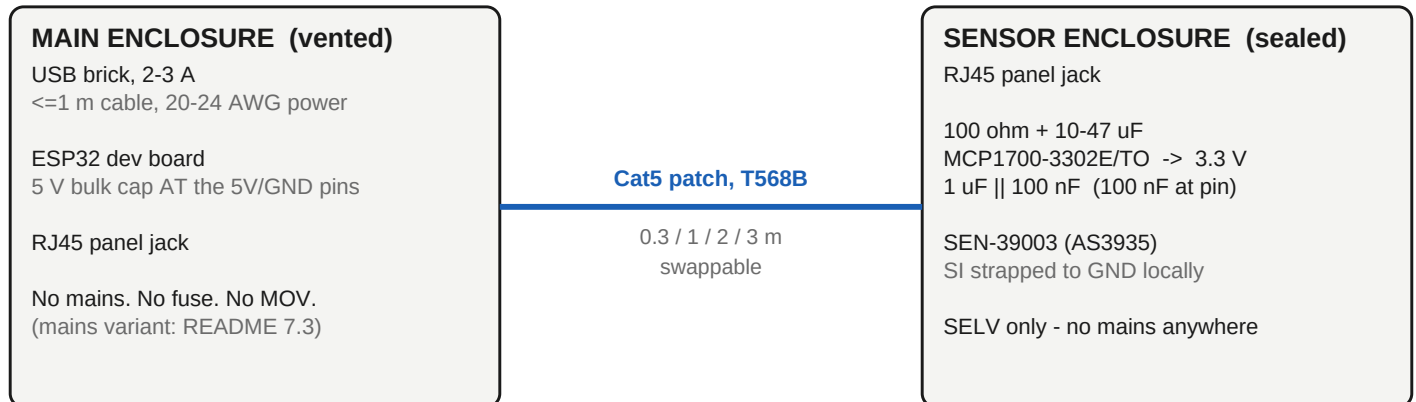


1. System overview

AS3935 lightning detector node - revision 2

Two enclosures. The patch cable length is a variable to be measured, not a fixed design choice - see README section 16.



Why two boxes

Near-field magnetic coupling falls as $1/r^3$, so 5 cm $\rightarrow$ 50 cm is roughly a 1000x reduction. Distance is the strongest lever available and nothing else is close. The interference source was never identified (README 11.3), so the separation is left adjustable on purpose: survey at each cable length and read the curve. A flat curve eliminates the ESP32 and the supply properly for the first time.

Do not plug this into a network switch

The RJ45 carries 5 V and SPI, not Ethernet. A live PoE port puts 48 V on those lines and destroys the ESP32 and the sensor. Label both ends.

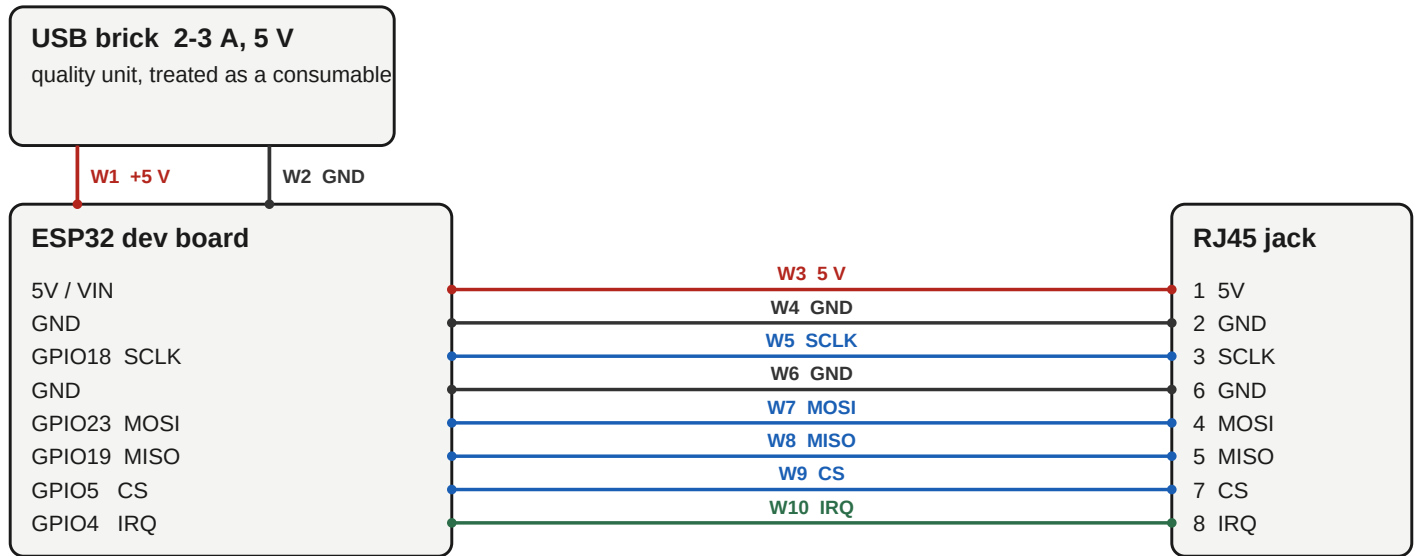
Signal assignment

T568B at both ends. Pairs are (1,2) (3,6) (4,5) (7,8). Seven signals plus one spare, and the spare is spent on a second ground paired with SCLK so the fastest edge gets its own return. SI is not carried - it straps to GND inside the sensor box.

2. Main enclosure

AS3935 lightning detector node - revision 2

USB supply, ESP32, and the RJ45 jack. Every line is one physical wire.



C1 470-1000 uF, 16-25 V, 105 C -- mounted AT the 5V/GND pins

Reservoir for the 300-500 mA WiFi bursts the cable resistance cannot supply fast enough.

The USB cable is a circuit element, not an accessory

28 AWG conductors are ~0.21 ohm/m. Over 2 m, counting the ground return, that is ~0.84 ohm: a 500 mA burst drops ~0.42 V, so 5.0 V arrives as 4.58 V. The onboard AMS1117 needs over a volt of headroom, so a thin or long cable browns the board out - the same failure the undersized IRM-02-5 produced, by a different route.

Use <=1 m with 20-24 AWG power conductors, and MEASURE at the 5V pin under WiFi load.

Keep the USB cable away from the Cat5 run; do not bundle them parallel.

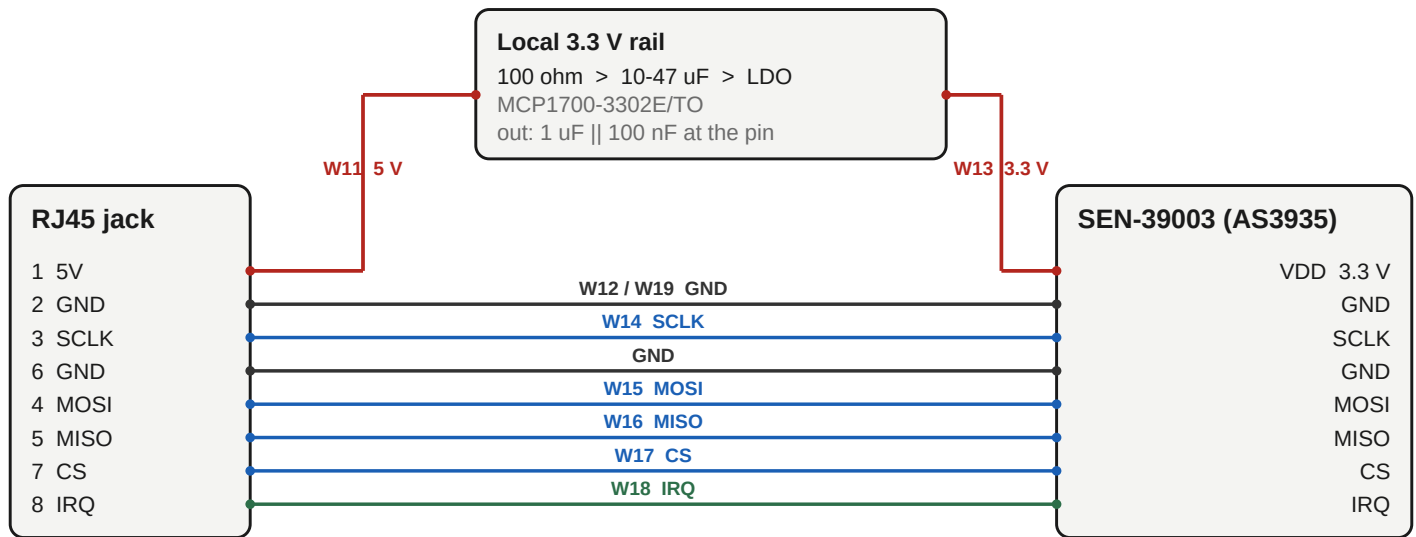
No mains in revision 2

The old rule - never have USB and the mains supply powered at the same time - does not apply, because there is only ever one supply. It returns in full if README 7.3 is built.

3. Sensor enclosure

AS3935 lightning detector node - revision 2

Everything here sits within a few centimetres of the AS3935. 5 V arrives on the cable; 3.3 V is made locally.



SI strapped to GND at the sensor board - selects SPI. Not carried on the cable.

Why a linear regulator, and why here

A switching regulator would put a 100 kHz - 1 MHz noise source centimetres from a 500 kHz magnetic antenna: the worst possible place for one. A linear regulator has no switching node, and at under 1 mA the wasted heat is nothing.

The E suffix is the -40/+125 C grade, required for a ~52 C attic. The 1 uF output cap is REQUIRED for MCP1700 stability, not optional.

The LDO does NOT replace the passives

LDO power-supply rejection is strong at low frequency - droop, WiFi burst sag - and is largely gone by 500 kHz. The LDO handles what the cable delivers at low frequency; the RC and the ceramics handle the band the AS3935 actually cares about. Neither suffices alone.

Set data_rate: 200kHz in the YAML

as3935_spi inherits the standard SPI device schema and DEFAULTS TO 1 MHz. At 200 kHz reflections over a few metres stop mattering; traffic is a few single-byte reads per event. IRQ is level-read in loop(), so cable capacitance cannot cost an interrupt.

4. Wire list and bring-up

AS3935 lightning detector node - revision 2

Wire numbers match pages 2 and 3.

#	From	To	Colour	Note
W1	USB brick +5 V	ESP32 5V / VIN	red	<=1 m, 20-24 AWG
W2	USB brick GND	ESP32 GND	black	with W1
W3	ESP32 5V	RJ45 pin 1	red	w/orange
W4	ESP32 GND	RJ45 pin 2	black	orange
W5	ESP32 GPIO18 SCLK	RJ45 pin 3	blue	w/green
W6	ESP32 GND	RJ45 pin 6	black	green - SCLK return
W7	ESP32 GPIO23 MOSI	RJ45 pin 4	blue	blue
W8	ESP32 GPIO19 MISO	RJ45 pin 5	blue	w/blue
W9	ESP32 GPIO5 CS	RJ45 pin 7	blue	w/brown
W10	ESP32 GPIO4 IRQ	RJ45 pin 8	green	brown
W11	RJ45 pin 1	100 ohm -> LDO in	red	sensor box
W12	RJ45 pins 2 + 6	LDO GND / star	black	sensor box
W13	LDO out 3.3 V	sensor VDD	red	caps at the pin
W14	RJ45 pin 3	sensor SCLK	blue	
W15	RJ45 pin 4	sensor MOSI	blue	
W16	RJ45 pin 5	sensor MISO	blue	
W17	RJ45 pin 7	sensor CS	blue	
W18	RJ45 pin 8	sensor IRQ	green	
W19	sensor GND	LDO GND / star	black	SI strapped here

Bring-up, in order

1. Assemble both boxes; connect with the SHORTEST patch cable.
2. Measure 5 V at the ESP32 5V pin DURING WiFi activity, not at idle. Want >4.7 V.
3. Measure 3.3 V at the sensor VDD pin, at the far end, after the LDO.
4. Verify tuning capacitance OVER SERIAL - the one check WiFi cannot do (README 12.1).
5. Confirm the sensor responds to the SEN-39002 emulator. Expect disturbers, not lightning.
6. Platform-validity test: survey, handle the build, survey again. Rates must not move.
7. Only then mount, and run the distance sweep at 0.3 / 1 / 2 / 3 m.

Step 6 is not optional

The previous breadboard build's interference floor fell by two thirds when it was touched, and nobody noticed for thirteen hours of apparently stable data. A platform that moves when handled cannot measure anything else. If the rates shift, fix the mechanics before going on.